The class diagram was introduced to represent a class in an object-oriented design of a solution, but it can be used to represent any class of object in a problem. You discover classes in your problem statement by underlining the nouns in the statement and making a list of these nouns. Then you examine the complexity of the nouns to decide whether a noun or object is complex enough to be a class or should be a structure within a class or a simple scalar variable.

A class diagram has three compartments, as shown in the template in figure A.4: the name of the class, the data area with a field and type for each data item, and the functions area. This template is from the draw.io tool, but you can create a class diagram using any UML tool or drawing package you are familiar with.

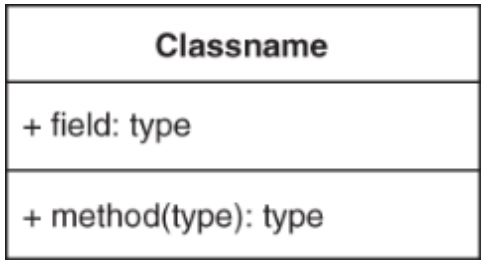



**Figure A.4 Class diagram template**

Let’s define a simple class diagram, using the automobile as an example.

> **PROBLEM STATEMENT** Design a class diagram representing a generic automobile.

Choose a simple name, `Auto`, for the class and then design the other parts of the class diagram. A rule of thumb for discovering data items is to ask and answer these questions: